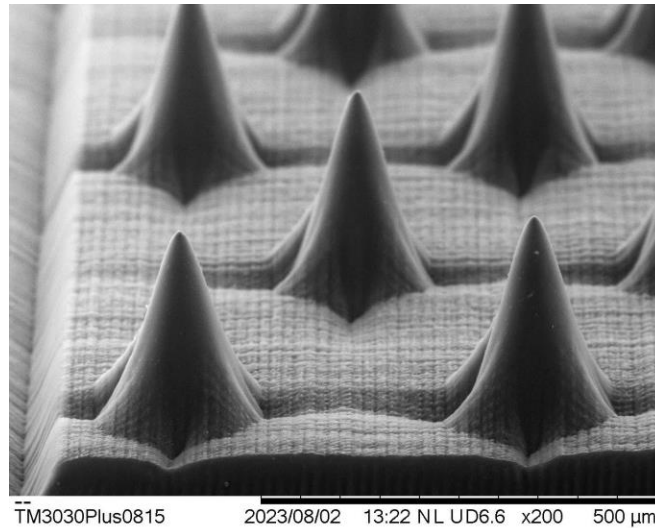


UROP End of Semester Essay – Ottavia Personeni

In the SPL Electrospray Thruster Fabrication project, my role mainly lies in the fabrication, testing, and characterization of ionic liquid ion sources for space propulsion. Over the course of the semester, I have prepared four complete thruster frames, ready to be etched. I will continue using this new porous material with 100-nm-scale pore size (Varapor 100). To my understanding, what is unique about the proposed tip structures is that the material is very well suited for extremely precise geometries, which can greatly improve electrospray performance.

Throughout the semester, I have cut many of these glass samples, using various methods to optimize the process. I noted that leaving glass samples placed in deionized water in petri dishes before and between cuts made the glass much easier to deal with. However, it was also crucial to lightly dry the samples in between cuts, since water from the glass cutting soaked the samples, making them weaker and more prone to cracks. I took notes to document procedures, successes, and failures.

I learned to prepare and set up thruster frames. I made four, carefully. I then sanded each of the glass pieces to align them exactly with the height of the frame and to polish them effectively. We noted a positive difference when using this new polishing technique. Hopefully, this will allow for more consistent etching and less deformations, which was the original goal. We will see how the magnitude of crust around each emitter will affect the probability of firing and the ability to transport propellant.



Over the course of the semester, many of these tasks had to be done carefully and slowly, to not harm limited supplies or small components, and to determine how to optimize each procedure step. Going forward, as I grow accustomed to cutting glass, building frames, and preparing mounts, and as I recognize and characterize the actions that maximize electrospray performance, this process will become more fluid.

Throughout my days in the SPL, I have also learned to use the VHX, such that when I begin conducting my own firing tests, I will be able to image thrusters before and after firing. Lastly, most recently, I was taught how to set up the frames in temporary mounts, and how to operate the TVAC. Matt explained what exactly happens in the ‘magic box,’ from a physics standpoint. I appreciated the explanation, as it gave me a sense for how the testing works - why it does what it does.

We practiced a testing procedure: gradually increasing the voltage applied until reaching startup voltage, as well as square wave and triangle wave firing. We discussed procedures for TVAC TOF scans and how the scans work, and the difficulties involved.

As an undergraduate researcher, my main duties this semester involved the manufacturing of the emitters. As we will soon begin testing and imaging and characterizing the thrusters built, I will see firsthand how adapting geometry and procedures optimizes performance. I plan to focus on developing a consistent and repeatable fabrication process, while making sure it remains well documented.

Next semester, we have plans to begin testing out unique propellants on the emitters. Since there exists preliminary evidence that by mixing lithium-based salts into regular ionic liquids, we can generate ion beams composed of almost 100% lithium ions, we hope to replicate this trial. This is a very useful undertaking that has significant implications, and I’m excited to get started in February.

My favorite part of my UROP work has proven to be the lab’s weekly seminar. It has been so interesting to see the research that each person does in the lab, how it overlaps with other projects, how it differs. Additionally, I love hearing Matt discuss work that I have helped out with. It all gives me an idea of the SPL community, and a sense of what it would be like to be part of a research-heavy graduate program. I also enjoyed hearing from Professor Yoshinori Takao as a testimony to the research that is being done around the world.

I really appreciate the dedication and knowledge Matt shows for his work. When he explains tasks or procedures, he tends to throw in cool bits of information from his own experiments or from a deeper physics knowledge. I take notes whenever we meet, and I like to write down these facts as well.

All the SPL research I have been exposed to has been intriguing. I am especially interested in seeing the thrusters fire. Since I have just been endowed with my own thruster to test and play around with, I will be able to collect firing data next semester. There are many applications of optimized electric propulsion techniques, and I am excited to learn more about electric propulsion as a whole as I continue taking aero astro classes and exploring the space propulsion industry. I am most looking forward to experimenting with unique propellants, hoping to replicate the 100% lithium-ion beam trial.

Through a continuation of this UROP, I look forward to gaining hands-on experience, particularly in studying how structural designs influence propulsion efficiency. I hope to actively

contribute to the characterization and optimized fabrication process. I also aim to continue my studies, as I am currently applying for graduate school, so I am excited to take advantage of all that this UROP has to offer, especially because I hope to explore novel ways to introduce electric propulsion to spacecraft.

Please see attached my notes for each procedure of this semester's SPL UROP. The documentation will be polished next semester.